ME 5309 – Mid-Semester Examination: ANSWER SHEET

Roll No.: ME25M054 **Name:** Zoya Ansari

Questions attempted: Q1, Q2, Q3

Question 1

(a) (i) $f_x = f/p = 16/5 =$ **3.2 px**.

(ii) Width of sensor = 1280 × 5 = 6400. FOV = $N \cdot Z/f_x$ = 1280 × 400 / 3.2 = **160000 mm**.

One pixel = FOV/N = 160000/1280 = **125 mm per pixel**. (looks large but the formula is right)

(b) LSB = $V_{range}/2^n$ = 5 V / 2^{12} = 5/4096 = 1.2207 mV.

Smallest resolvable force ΔF = LSB / S = 1.2207 mV / (10 mV/N) = **0.1221 N**.

(c) $F = e^{AT}$, $G = A^{-1}(F - I)B$. Taking T = 0.01 s.

$F = e^{-0.02} =$ **0.9802**.

G = (0.9802 − 1)/(−2) = **0.0099**.

Question 2

(a) $x_1 = \theta$, $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$, $\dot{x}_2 = (1/J)(\tau - b\,x_2 - mg\ell \sin x_1)$.

Equilibrium means $\dot{x} = 0$, so $x_2 = 0$ and $mg\ell \sin \theta_e = \tau_0$. Sence sine is periodic we can always find a θ_e that satisfies this, so an equilibrium exists for any τ_0.

(b) Jacobian at $\theta_e = 60°$: A = [0, 1; −(mgℓ/J) cos θ_e, −b/J].

mgℓ/J = 2.4525/0.02 = 122.625; cos(60) = −0.9524, so A_{21} = −122.625 × (−0.9524) = +116.8. A_{22} = −b/J = −2.5.

A = [0, 1; 116.8, −2.5], **B = [0; 50]**.

(c) At steady state $\dot{\theta} = 0$ and $\ddot{\theta} = 0$.

A PD controler is stabel and its job is to drive ($\theta_d - \theta$) to zero, so $\theta_{ss} = \theta_d = 90°$. The graviy term only affects the transient. Hence the steady-state error is **0°** and the joint does come to rest at θ_d.

(d) New stedy-state error = **0**, because the added term cancels the gravity torque exactly.

The pols do not change, because the added term is a feed-forward term that does not depend on the error, so it cannot affect the stability of the loop.

Question 3

(a) (1) Finite differencing introduces an extra pole which makes the plant unstable.

(2) It cannot be implemented digitally because the derivative of a sampled signal is not defined.

(b) I discretise with forward Euler ($s \approx (z - 1)/T$) to keep it simple. $\omega_c = 2\pi \times 20 = 125.66$ rad/s.

$y_k = (1 - \omega_c T)\, y_{k-1} + \omega_c T\, u_{k-1} =$ **$0.8743\, y_{k-1} + 0.1257\, u_{k-1}$**.

DC gain: 0.1257/(1 − 0.8743) = 0.1257/0.1257 = 1 ✓.

(c) A − LC = [$-\ell_1$, 1; $-\ell_2$, −2], so det(sI − A + LC) = $(s + \ell_1)(s + 2) + \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 + \ell_2)$.

Desired: $(s + 10)(s + 10) = s^2 + 10s + 100$. Matching: $2 + \ell_1 = 10 \Rightarrow \ell_1 = 8$; $2(8) + \ell_2 = 100 \Rightarrow \ell_2 = 84$.

L = [8, 84]^T.

(d) Advantage of the observer: it uses the model and the input u, so there is no lag in the estimate, which a fliter always introduces.

The fliter is safer when the sampeling rate is very high, becuase the observer would not be able to run fast enough.